

Mathematics for Machine Learning

<https://mml-book.github.io/book/mml-book.pdf>

1 Linear Algebra

1.1 vectors and systems

- Polynomials form a vector space.
- Elements of $\mathbb{R}^n$ (tuples of n real numbers) are vectors. For example,

$$a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^3.$$

- Adding two vectors $a, b \in \mathbb{R}^n$ component-wise results in another vector:
 $a + b = c \in \mathbb{R}^n$.
- Scaling: $a \in \mathbb{R}^n, \lambda \in \mathbb{R} \implies \lambda a \in \mathbb{R}^n$.
- Systems of equations: solution by contradiction,

$$x_1 + x_2 + x_3 = 3 \tag{1}$$

$$x_1 - x_2 + 2x_3 = 2 \tag{2}$$

$$2x_1 + 3x_3 = 1 \tag{3}$$

has *no solution*: Adding the first two equations yields $2x_1 + 3x_3 = 5$, which contradicts the third equation (3).

- Systems of equations: idk this is kinda basic but still,

$$x_1 + x_2 + x_3 = 3 \tag{1}$$

$$x_1 - x_2 + 2x_3 = 2 \tag{2}$$

$$x_2 + x_3 = 2 \tag{3}$$

$$(1) + (2) = 2x_1 + 3x_3 = 5, \text{ i.e., } x_3 = 1.$$

From (3): we then get that $x_2 = 1$. so $(1, 1, 1)$ is the only possible and *unique solution*

- Systems of equations: omission

$$x_1 + x_2 + x_3 = 3 \tag{1}$$

$$x_1 - x_2 + 2x_3 = 2 \tag{2}$$

$$2x_1 + 3x_3 = 5 \tag{3}$$

Since $(1) + (2) = (3)$, we omit the third equation. solve with (1) and (2) and let $x_3 = a \in \mathbb{R}$.